YEAR 10 SCIENCE EXTENSION
NATURAL AND PROCESSED MATERIALS
ASSIGNMENT 2: CHEMICAL CALCULATIONS

NAME: ____ **Total:** $\frac{\quad}{34}$

1. Calculate the formula mass for the following substances.

NAME	FORMULA	FORMULA MASS ($gmol^{-1}$)
sodium phosphate		
iron (III) nitrate		
calcium hydrogencarbonate		
barium sulphide		
zinc acetate		

(10)

2. Convert the following to moles or mass as necessary.

(a) 36.2 g of copper (II) nitrate.

(b) 2.34 moles of aluminium hydroxide.

(b) 1.34 kg of iron (III) oxide.

(d) 1.15×10^3 moles of sulphur trioxide.

(8)

3. 27.2 g of aluminium carbonate is reacted with excess sulphuric acid. The equation for the reaction is as follows.

$$Al_2(CO_3)_{3(s)} + 3\ H_2SO_{4(aq)} \rightarrow Al_2(SO_4)_{3(aq)} + 3\ H_2O_{(l)} + 3\ CO_{2(g)}$$

(a) Calculate the mass of sulphuric acid used in the reaction?

(5)

(b) What mass of carbon dioxide would be produced?

(3)

4. Ethanol burns in air according to the following equation.

$$C_2H_5OH_{(l)} + 3\ O_{2(g)} \rightarrow 2\ CO_{2(g)} + 3\ H_2O_{(g)}$$

If 3.85×10^2 g of ethanol burns, calculate:

(a) the mass of O_2 required.

(5)

(b) the mass of CO_2 produced.

(3)